A SIXTH-DEGREE MIXING BOUND FOR THE FOUR-VERTEX SWITCH CHAIN

Abstract. For every $n \geq 4$ and every graphical labeled degree vector d, the lazy four-vertex-proposal switch chain has total-variation mixing time at distance $1/4$ at most n^6. The proposal rule is exactly that of the September 25, 2026 OpenAI manuscript, whose bound is $2n^8$. The argument retains its pair-resampling Poincaré inequality and uses the subset-slice estimate $J \succeq N(I - E)$ to obtain a switch-Laplacian gap of at least one. All needed arguments are included, with the inherited proof distinguished from the sharper comparison. The note records the complete argument and its improvement over the original OpenAI bound.

1. The model, result, and attribution

Let $V = [n]$, $n \geq 4$, and let $d = (d_1, \dots, d_n)$ be any graphical labeled degree vector. Let Ω_d be the nonempty set of simple undirected graphs on V with degrees d, and let π be its uniform law. There is no restriction on the maximum degree or the number of edges. A switch removes two vertex-disjoint present edges and inserts a different perfect matching on the same four vertices, provided the two inserted edges are absent.

The discrete-time kernel P_d first holds with probability $1/2$. Otherwise it selects a uniformly random four-element set $S \subseteq V$ and a uniformly random ordered pair (F, F') of distinct perfect matchings on S, among the six possibilities. If both edges of F are present and both edges of F' are absent, it replaces F by F'; otherwise it holds. This is the exact convention of [2], with rejected proposals counted as steps. Define

$$\|\mu - \nu\|_{\mathrm{TV}} = \frac{1}{2} \sum_{G \in \Omega_d} |\mu(G) - \nu(G)|,$$

$$t_{\mathrm{mix}}(d; 1/4) = \min\{t \in \mathbb{Z}_{\geq 0} : \max_G \|P_d^t(G, \cdot) - \pi\|_{\mathrm{TV}} \leq 1/4\}.$$

Theorem 1.1. *For every $n \geq 4$ and every graphical labeled degree vector d,*

$$t_{\mathrm{mix}}(d; 1/4) \leq n^6.$$

If $|\Omega_d| > 1$, the spectral gap of P_d is at least $[12\binom{n}{4}]^{-1}$. Thus the constants in the accepted statement $t_{\mathrm{mix}}(d; 1/4) \leq Cn^p$, $C > 0$, $p < 8$, can be chosen as $C = 1$, $p = 6$.

The inherited argument is the pair-resampling proof in [2, Sections 2–7], including the three-row estimate, disjoint-pair calculation, and connectivity. Appendix A reproduces the admitted dependency proof. The three-row

Date: October 17, 2026.

variance inequality and adjacent-total coupling are credited in that source to Fu, Qin, and Wang [1]. The sharper step used here replaces the source's comparison $h_a \preceq n^2 J_a$ by $h_a \preceq J_a/2$. The subset-slice spectrum is the classical input used here. The complete proof components are reproduced below.

2. Pair resampling and the subset-slice estimate

All inner products are real L^2 inner products for the relevant uniform law. The symbol $A \succeq B$ denotes quadratic-form order. For an unordered pair $a = \{i, j\}$, order its endpoints and fix all edges outside a, the edge ij, and, for every $w \notin a$, the number of its neighbors in a. These data define an a-fiber. Its singleton vertices are the N outside vertices with exactly one neighbor in a. The degree of i fixes the number q of singletons assigned to i; the fiber is precisely the uniform q-subset slice on these N labels. Let E_a average on fibers, $h_a = I - E_a$, and $H = \sum_{a \in \binom{V}{2}} h_a$. The admitted prerequisite, proved in Appendix A, is

$$\operatorname{Var}_\pi(f) \le \langle f, Hf \rangle. \tag{2.1}$$

Lemma 2.1. *On the uniform slice $X = \{x \in \{0,1\}^N : \sum_i x_i = q\}$, let τ_{ij} exchange coordinates, $J = \sum_{i<j}(I - \tau_{ij})$, and E average over X. If $1 \le q \le N-1$, then*

$$J \succeq N(I - E). \tag{2.2}$$

For $q = 0, N$, both J and $I - E$ vanish.

Proof. For $A \subseteq [N]$, put $x_A = \prod_{i \in A} x_i$, with $x_\varnothing = 1$, and let U_k be the span of the monomials of degree at most k on the slice. For $|A| = k \ge 1$, only swaps with exactly one index in A contribute. Using $x_i^2 = x_i$ and $\sum_i x_i = q$ gives

$$\begin{aligned} Jx_A &= \sum_{i \in A,\, j \notin A} (x_A - x_{A \setminus \{i\}} x_j) \\ &= k(N-k+1)x_A - (q-k+1)\sum_{i \in A} x_{A \setminus \{i\}}. \end{aligned} \tag{2.3}$$

Consequently U_k is invariant, and J acts by $\lambda_k = k(N-k+1)$ modulo U_{k-1}. Self-adjointness makes $W_k = U_k \cap U_{k-1}^{\perp}$ invariant. For $w \in W_k$, $(J - \lambda_k I)w$ lies in both U_{k-1} and W_k, so it vanishes. The degree-q monomials are the point indicators of q-subsets, hence U_q is the full function space. It is the orthogonal sum of the constants and $W_1, \dots, W_q$, allowing zero summands. For $1 \le k \le q \le N-1$,

$$\lambda_k - N = (k-1)(N-k) \ge 0.$$

Since J kills constants, this proves (2.2). Singleton slices give the other cases, including $N = 0, 1$. □

3. The switch comparison and the exact proposal clock

Let L be the unnormalized Laplacian of the simple switch graph,

$$(Lf)(G) = \sum_{G' \text{ a distinct switch neighbor of } G} (f(G) - f(G')).$$

On each a-fiber define J_a by the J of Lemma 2.1, and combine the operators over fibers. Every nonconstant fiber has $N \geq 2$, so $h_a \preceq J_a/N \preceq J_a/2$ there; both operators vanish on constant fibers. Thus globally

$$h_a \preceq \tfrac{1}{2} J_a, \qquad \sum_a J_a = 2L, \qquad H \preceq L. \tag{3.1}$$

For the middle identity, a singleton transposition toggles a four-cycle whose bipartition classes are the indexing pair and the exchanged singleton pair. A switch $uv, xy \longmapsto ux, vy$ has exactly two such representations, with indexing pairs $\{u, y\}$ and $\{v, x\}$. Each determines the transposition uniquely; no other pair represents the same toggled cycle. Transpositions of identical assignments contribute zero. Both the off-diagonal entries and diagonal exit contributions are therefore counted twice. Combining (2.1) and (3.1),

$$\mathrm{Var}_\pi(f) \leq \langle f, Hf \rangle \leq \langle f, Lf \rangle. \tag{3.2}$$

For a state space with more than one graph, L has gap at least one.

Every distinct switch neighbor uniquely determines the four-set S, removed matching F, and added matching F'. Thus

$$P_d(G, G') = \frac{1}{12\binom{n}{4}}, \qquad I - P_d = \frac{L}{12\binom{n}{4}}. \tag{3.3}$$

The reverse probability is the same, proving reversibility for π. Write $P_d = (I + Q)/2$, where Q includes rejected proposals. Jensen's inequality and stationarity give $\|Q\|_{2\to 2} \leq 1$; reversibility makes Q self-adjoint. All eigenvalues of P_d are nonnegative. Equations (3.2)–(3.3) put its centered eigenvalues in $[0, 1 - \gamma]$, where

$$\gamma = \frac{1}{12\binom{n}{4}}. \tag{3.4}$$

Proof of Theorem 1.1. If $M = |\Omega_d| = 1$, the mixing time is zero. Otherwise the density of a point mass at G relative to π, minus one, is centered with squared L^2 norm $M - 1$. Reversibility evolves this density by P_d. The spectral estimate and Cauchy–Schwarz give, for integer $t \geq 0$,

$$\|P_d^t(G, \cdot) - \pi\|_{\mathrm{TV}} \leq \tfrac{1}{2}\sqrt{M-1}(1-\gamma)^t \leq \tfrac{1}{2}\sqrt{M}\exp(-\gamma t).$$

Consequently distance $1/4$ is reached by

$$t = \left\lceil 12\binom{n}{4}\left(\log 2 + \tfrac{1}{2}\log M\right)\right\rceil. \tag{3.5}$$

All logarithms are natural. There are at most $2^{\binom{n}{2}}$ labeled simple graphs, so, for $n \geq 4$,

$$\log 2 + \tfrac{1}{2}\log M \leq \left(1 + \frac{n(n-1)}{4}\right)\log 2 \leq n^2, \qquad 12\binom{n}{4} \leq \frac{n^4}{2}.$$

Thus (3.5) is at most $\lceil n^6/2 \rceil \leq n^6$. The gap conclusion follows from (3.4). □

4. Comparison with the original bound

The pinned September 25, 2026 version of [2, Theorem 1.1] states $2n^8$ and gap $[24n^2\binom{n}{4}]^{-1}$ for exactly the chain above. The bound proved here satisfies $n^6/(2n^8) = 1/(2n^2) < 1$ for every $n \geq 4$. It improves the polynomial exponent from eight to six while retaining all graphical degree vectors and the original proposal rule.

The result concerns simple undirected graphs, the exact lazy four-vertex proposal kernel and total-variation distance $1/4$.

Appendix A. The inherited pair-resampling Poincaré proof

This appendix is an editorial transcription of the admitted predecessor proof. Its mathematical argument is attributed to [2, Sections 2–7]; the three-row estimate and adjacent-total coupling retain the attribution to [1]. It proves (2.1) without assuming an unverified source assertion.

A.1. **The three-row variance inequality.** First take independent uniform ordered partitions (U_1, U_2, U_3) and (W_1, W_2, W_3) of two labeled sets, with all block sizes fixed. For real functions $g_i(U_i, W_i)$,

$$\operatorname{Var}\left(\sum_i g_i\right) \leq 2\sum_i \operatorname{Var}(g_i). \tag{A.1}$$

To prove this, center each g_i and let $\mathcal{J} = \sum_T (I - T)$, summing over transpositions of labels within either underlying set. Each T is a self-adjoint involution, and $\langle f, \mathcal{J} f\rangle = \frac{1}{2}\sum_T \mathbb{E}[(f - Tf)^2]$. Transpositions act transitively, so the kernel is the constants. The functions of (U_i, W_i) form an invariant space: the effect of a transposition on block i is determined solely by that block and the two labels. Spectral projections, being polynomials in $\mathcal{J}$, preserve this space. At a positive eigenvalue λ, write the components as $g_{i,\lambda}$. A transposition changes at most two block tuples, so pointwise

$$\left(\sum_i (g_{i,\lambda} - T g_{i,\lambda})\right)^2 \leq 2\sum_i (g_{i,\lambda} - T g_{i,\lambda})^2.$$

Sum the energy identity, divide by λ, and sum over orthogonal eigenspaces to obtain (A.1). Empty blocks and singleton partition spaces are included.

Next, let $x_1 + x_2 + x_3 = m$ have law proportional to $\prod_i w_i(x_i)$ on nonnegative integers, where $w_i(t) = [t!(t+\alpha_i)!]^{-1}$ and $\alpha_i \geq 0$ is integral. Then

$$\operatorname{Var}\left(\sum_i b_i(x_i)\right) \leq 2\sum_i \operatorname{Var}(b_i(x_i)). \tag{A.2}$$

For two coordinates j, k of total s, call the first Y_s and the second $Z_s = s - Y_s$. On $0 \leq y \leq s-1$, the ratio of the probabilities for Y_s and Y_{s-1} is a positive constant times $[(s-y)(s-y+\alpha_k)]^{-1}$, increasing in y. Reweighting by an increasing function stochastically increases a real variable, since for increasing a, b and independent copies Y, Y',

$$\operatorname{Cov}(a(Y), b(Y)) = \tfrac{1}{2}\mathbb{E}[(a(Y) - a(Y'))(b(Y) - b(Y'))] \geq 0.$$

Including mass at s gives $Y_{s-1} \leq_{\mathrm{st}} Y_s$. Interchanging j, k gives $Z_{s-1} \leq_{\mathrm{st}} Z_s$, or $Y_s \leq_{\mathrm{st}} Y_{s-1} + 1$. Common quantiles couple these simultaneously: $Y_s - Y_{s-1} \in \{0, 1\}$, and exactly one coordinate increases by one.

For (A.2), $m = 0$ is immediate. Map the direct sum of the three centered marginal L^2 spaces to the centered joint space by $A(b) = \sum_i b_i(x_i)$. A nonzero eigenvector of A^*A for its maximal eigenvalue λ satisfies

$$\lambda b_i(t) = \mathbb{E}\left[\sum_j b_j(x_j) \mid x_i = t\right], \qquad 0 \leq t \leq m.$$

Every marginal probability is positive. The maximum

$$M_0 = \max_{i,\, 0 \leq t < m} |b_i(t+1) - b_i(t)|$$

is positive, since otherwise all centered components vanish. Subtract the equation at adjacent t. The other two coordinates have adjacent totals, and the coupling above changes exactly one coordinate by one. Thus $|(\lambda - 1)(b_i(t+1) - b_i(t))| \leq M_0$. An increment attaining M_0 gives $|\lambda - 1| \leq 1$, hence $\lambda \leq 2$, proving (A.2).

Now take a uniform nonempty space of three-row binary matrices with fixed row and column sums. Remove deterministic columns of sum zero or three. Partition the labels of sum-one columns into U_i according to their unique one, and those of sum-two columns into W_i according to their unique zero. Row i is specified by (U_i, W_i). With $k_i = |W_i|$, fixed integers δ_i satisfy $|U_i| = k_i + \delta_i$. Given k, these are independent uniform partitions of prescribed sizes; the count law is proportional to

$$\prod_i \frac{1}{k_i!(k_i+\delta_i)!}, \qquad k_i \geq 0, \quad k_i + \delta_i \geq 0, \quad \sum_i k_i = |W|.$$

These constraints suffice because both lists of block sizes have the required totals. Set $l_i = \max(0, -\delta_i)$, $x_i = k_i - l_i$, $\alpha_i = |\delta_i|$, and $m = |W| - \sum_i l_i \geq 0$. This is the count law above. For a row function f_i, $\mathbb{E}[f_i \mid k]$ depends only on k_i, because its two blocks are independent uniform subsets of their prescribed

sizes. Conditional variance, (A.1), and (A.2) imply

$$\operatorname{Var}\left(\sum_i f_i(\mathrm{row}_i)\right) \le 2\sum_i \operatorname{Var}(f_i(\mathrm{row}_i)). \tag{A.3}$$

A.2. **Overlapping pair updates.** Fix a triple $T \subseteq V$. Partition Ω_d by the graph on $V \setminus T$ and each outside vertex's number of neighbors in T. The pair resamplings within T preserve these cells. In a cell the number e of internal edges is fixed by the total degree. Use the three-row matrix of edges from T to outside vertices. For $e = 0, 3$, the internal graph is fixed, with row sums d_i or $d_i - 2$. For $e = 1$, append the internal-degree column s_i, of sum two, with row sums d_i. For $e = 2$, append $s_i - 1$, of sum one, with row sums $d_i - 1$. The added column records the sole edge's endpoints or the path's center, respectively. These are bijections onto the full binary matrix spaces of the specified margins, preserving uniform measure.

For $i \in T$, write $\{j, k\} = T \setminus \{i\}$. Conditioning on row i is exactly conditioning on the $\{j, k\}$-fiber: the external row determines s_i from d_i, its added entry, and all external edges at i; the edge jk has indicator $e - s_i$. An outside vertex's count in $\{j, k\}$ is its fixed T-count minus its known edge to i, and i's count is s_i. These are precisely the fiber data; the converse holds as well. Let P_i be the row conditional expectations. For cell-centered f, put $b = \sum_i \|P_i f\|^2$. By (A.3) and Cauchy–Schwarz,

$$b = \langle f, \sum_i P_i f\rangle \le \|f\|\sqrt{2b}, \qquad b \le 2\|f\|^2.$$

Thus $H_T = \sum_{a \in \binom{T}{2}} h_a = 3I - \sum_i P_i$ kills cell constants and has every nonzero eigenvalue at least one. It follows cellwise, and then globally, that

$$H_T^2 \succeq H_T. \tag{A.4}$$

A.3. **Equality projections on a slice.** On a uniform q-subset slice on $[N]$, put $\mathcal{G} = \frac{1}{2}\sum_{i<j}(I - \tau_{ij})$ and $z_{ij} = \mathbf{1}_{\{x_i = x_j\}} - \Pr(x_i = x_j)$. Let Q_{ij} project onto the span of z_{ij}, or be zero if it vanishes. The required inequality is

$$S := \sum_{i<j} Q_{ij} \preceq \mathcal{G}. \tag{A.5}$$

If all z_{ij} vanish the claim is immediate, including $N \le 2$ and $q = 0, N$. Otherwise $N \ge 3$, $1 \le q \le N - 1$. Set $r = N - q$ and

$$p = \frac{q(q-1) + r(r-1)}{N(N-1)}, \qquad 1 - p = \frac{2qr}{N(N-1)}, \qquad \|z_{ij}\|^2 = p(1-p) > 0.$$

On the Euclidean coefficient space of $M_1 = \binom{N}{2}$ pairs, define

$$Bc = \sum_{i<j} \frac{c_{ij} z_{ij}}{\sqrt{p(1-p)}}.$$

Then $S = BB^*$ and $K = B^*B$ is positive semidefinite. Decompose coefficients orthogonally into C_0, the constants; $C_1 = \{c_{ij} = u_i + u_j : \sum_i u_i = 0\}$; and $C_2 = \{\sum_{j \ne i} c_{ij} = 0 \text{ for every } i\}$. For $N \ge 3$, the map $u \mapsto (u_i + u_j)$ is

injective by considering a triangle. The dimensions are $1, N-1, M_1-N$; $C_2 = 0$ when $N = 3$.

The number of equal pairs is fixed, so B kills C_0. By permutation invariance, entries of K depend only on whether their indexing pairs coincide, intersect once, or are disjoint. For $c \in C_1$, the sums over other pairs meeting ij and over disjoint pairs are $(N-4)c_{ij}$ and $-(N-3)c_{ij}$; for $c \in C_2$ they are $-2c_{ij}$ and c_{ij}. These follow by adding row sums and using the zero total sum. Hence K acts by scalars μ_1, μ_2 on these two spaces. The identity $\sum_{j\neq i} \mathbf{1}_{\{x_i=x_j\}} = (r-1) + (2q-N)x_i$ gives

$$\sum_{i<j}(u_i+u_j)z_{ij} = (2q-N)\sum_i u_i x_i.$$

Since $\mathbb{E}x_i = q/N$ and $\mathbb{E}x_ix_j = q(q-1)/[N(N-1)]$,

$$\operatorname{Var}\left(\sum_i u_i x_i\right) = \frac{qr\sum_i u_i^2}{N(N-1)}, \qquad \sum_{i<j}(u_i+u_j)^2 = (N-2)\sum_i u_i^2.$$

Therefore

$$\mu_1 = \frac{(2q-N)^2}{2(N-2)p} \le \frac{N}{2},$$

where the inequality follows from $N(N-2)p-(2q-N)^2 = \frac{2N}{N-1}[qr-(N-1)] \ge 0$. The diagonal of K is one, so $\operatorname{tr} K = M_1$. For $N \ge 4$, positivity gives $0 \le \mu_2 \le M_1/(M_1-N) = (N-1)/(N-3) \le N-1$.

The orthogonal images BC_i carry S with eigenvalues μ_i. Direct transposition sums give

$$\mathcal{G}x_i = \tfrac{1}{2}(Nx_i - q), \qquad \mathcal{G}(x_ix_j) = (N-1)x_ix_j - \tfrac{1}{2}(q-1)(x_i+x_j).$$

Thus $\mathcal{G}$ acts by $N/2$ on BC_1. For $c \in C_2$, the constant and linear terms in $\sum c_{ij}z_{ij}$ vanish, leaving $2\sum c_{ij}x_ix_j$; its lower-degree terms under $\mathcal{G}$ also vanish by the zero row sums. Hence the eigenvalue on BC_2 is $N-1$. The images are orthogonal since K is scalar on the orthogonal coefficient spaces. The scalar comparisons prove (A.5) on their sum. Its orthogonal complement is $\mathcal{G}$-invariant, with $S = 0$ and $\mathcal{G} \succeq 0$, completing the proof, including zero images and the case $N = 3$.

A.4. **The disjoint-pair calculation.** For disjoint pairs a, b, let T_{ab} toggle the four cross edges when their 2×2 matrix has every row and column sum one, and otherwise be the identity. This is a measure-preserving involution; $g_{ab} = (I - T_{ab})/2 = g_{ba}$ is an orthogonal projection. In an a-fiber, let R_{ab} project onto the centered equality indicator for the assignments of the two vertices in b; set it to zero if either is not a singleton or the centered indicator vanishes. Take direct sums over fibers. The local estimate is

$$2\langle h_a f, h_b f\rangle \ge 2\|g_{ab}f\|^2 - \|R_{ab}f\|^2 - \|R_{ba}f\|^2. \tag{A.6}$$

Write $a = \{i,j\}$, $b = \{k,l\}$, and $O = V \setminus (a \cup b)$. Partition by the graph on O, edges ij, kl, and the separate neighbor counts in a and b of each $w \in O$. These cells are invariant under both updates. A state in a cell is described by

the cross matrix M, a subset U of the outside singleton set A_0 for a assigned to i, and a subset W of the corresponding set B_0 for b assigned to k. Let s be the total of M, r its first-row sum, and c its first-column sum. The constraints are exactly

$$|U| = q_0 - r, \qquad |W| = q_1 - c$$

for fixed integers q_0, q_1, with s fixed. Every binary M of total s and every feasible subset pair has equal probability. An intersection of A_0 and B_0 imposes no further constraint, since it concerns distinct edges. The complete fiber data are $X = (c, W)$ for a and $Y = (r, U)$ for b. Denote their function spaces by $\mathcal{A}, \mathcal{B}$ and put $\mathcal{J}_0 = (\mathcal{A} + \mathcal{B})^{\perp}$. Then $E_a = \mathbb{E}[\cdot \mid X]$, $E_b = \mathbb{E}[\cdot \mid Y]$. Since T_{ab} fixes X, Y, $\operatorname{ran} g_{ab} \subseteq \mathcal{J}_0$.

For $s = 0, 4$, the matrix is fixed; for $s = 1, 3$, its unique one or zero is determined separately by the row and column sums. The variables X, Y are independent in these cases. For $s = 2$, the multiplicities are

$$(m_{rc})_{r,c=0}^{2} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The events $D_r = \{r \neq 1\}$ and $D_c = \{c \neq 1\}$ are disjoint. If both have positive probability, the central state is feasible: D_r supplies a feasible W of size $q_1 - 1$ and D_c supplies a feasible U of size $q_0 - 1$, together realizing both central matrices. If an event has probability zero, or the center is absent, the support lies in a single row or column. Matrix multiplicity then depends on only one coordinate, and subset counts factor, so X, Y are independent. In every independent case $E_a E_b = E_b E_a$ projects onto cell constants, $h_a h_b$ projects onto $\mathcal{J}_0$, and $2\langle h_a f, h_b f\rangle = 2\| \operatorname{proj}_{\mathcal{J}_0} f\|^2 \geq 2\|g_{ab} f\|^2$, proving (A.6).

It remains that $s = 2$ and both events and the center have positive probability. Put $p_r = \Pr(D_r)$, $p_c = \Pr(D_c)$, and

$$u = \frac{\mathbf{1}_{D_c} - p_c}{\sqrt{p_c(1-p_c)}}, \quad v = \frac{\mathbf{1}_{D_r} - p_r}{\sqrt{p_r(1-p_r)}}, \quad \rho = \sqrt{\frac{p_c p_r}{(1-p_c)(1-p_r)}}.$$

Then $\langle u, v\rangle = -\rho$ and $0 < \rho < 1$. For each feasible U,

$$\Pr(Y = (r, U) \mid X = (c, W)) = \frac{m_{rc}}{\sum_{t=0}^{2} \binom{|A_0|}{q_0 - t} m_{tc}},$$

with infeasible binomial coefficients interpreted as zero. This law depends only on whether $c = 1$; symmetrically the reverse law depends only on whether $r = 1$. It follows that $E_a \mathcal{B} \subseteq \operatorname{span}(1, u)$, $E_b \mathcal{A} \subseteq \operatorname{span}(1, v)$, $E_a v = -\rho u$, and $E_b u = -\rho v$.

Let $\mathcal{A}'$ be the centered functions in $\mathcal{A}$ orthogonal to u, and define $\mathcal{B}'$ analogously. These identities give $\mathcal{A}' \perp \mathcal{B}$ and $\mathcal{B}' \perp \mathcal{A}$. The cell space is the orthogonal sum of constants, $\mathcal{A}'$, $\mathcal{B}'$, $\operatorname{span}(u, v)$, and $\mathcal{J}_0$; each summand is invariant under E_a, E_b. The first three contribute zero to $\langle h_a f, h_b f\rangle$ and the last contributes $\| \operatorname{proj}_{\mathcal{J}_0} f\|^2$. For a plane component $xu + yv$, its h_a, h_b

images are $y(v+\rho u)$ and $x(u+\rho v)$. The two parenthesized vectors have squared norm $1-\rho^2$ and inner product $\rho(1-\rho^2)$, giving

$$2xy\rho(1-\rho^2) \geq -(x^2+y^2)(1-\rho^2).$$

Here $2|xy| \leq x^2+y^2$ and $\rho < 1$.

The vector $y(v+\rho u)$ is the projection of the full f onto the line spanned by $v+\rho u$, since that line is perpendicular to u and every other summand. The line lies in $\operatorname{ran} R_{ab}$. On an X-fiber with $c \neq 1$ it vanishes; if $c=1$, the vertices k, l are singletons for a and D_r is their equality event, so

$$v - E_a v = \frac{\mathbf{1}_{\text{equal}} - \Pr(\text{equal} \mid X)}{\sqrt{p_r(1-p_r)}}.$$

The other line lies in $\operatorname{ran} R_{ba}$ similarly. Thus $\|y(v+\rho u)\| \leq \|R_{ab}f\|$ and $\|x(u+\rho v)\| \leq \|R_{ba}f\|$. Together with $\operatorname{ran} g_{ab} \subseteq \mathcal{J}_0$ this proves (A.6) in the cell. The operators, including R_{ab}, preserve cells, so averaging proves it globally.

A.5. **Summation, connectivity, and the kernel.** On an a-fiber, when b consists of singleton labels k, l, $T_{ab} = \tau_{kl}$ and $R_{ab} = Q_{kl}$ from Section A.3. If either is not a singleton, both g_{ab} and R_{ab} vanish. Since a projection Q satisfies $\|Qf\|^2 = \langle f, Qf\rangle$, (A.5) gives

$$\sum_{b:\, b\cap a=\varnothing} \|R_{ab}f\|^2 \leq \sum_{b:\, b\cap a=\varnothing} \|g_{ab}f\|^2. \tag{A.7}$$

Let $D = \sum_{a,b:\, a\cap b=\varnothing} h_a h_b$, with ordered indices. Both orders occur, so D is self-adjoint. Grouping unordered pairs in (A.6) and using $g_{ab} = g_{ba}$ yields

$$\langle f, Df\rangle \geq \sum_a \sum_{b:\, b\cap a=\varnothing} (\|g_{ab}f\|^2 - \|R_{ab}f\|^2) \geq 0.$$

Expansion by overlapping and disjoint indices, using $h_a^2 = h_a$, gives

$$H^2 = \sum_{T\in\binom{V}{3}} H_T^2 - (n-3)H + D.$$

Each ordered overlapping pair occurs in its unique triple, while each h_a occurs $n-2$ times in the local squares. By (A.4),

$$H^2 \succeq (n-2)H - (n-3)H = H. \tag{A.8}$$

For completeness, freeze vertices one at a time in label order to prove switch connectivity. On the remaining set R, with current residual degrees d_i^R, prescribe the neighbors of its first vertex v to be the d_v^R vertices of greatest residual degree, breaking ties by label. If a desired j is missing and an undesired i is present, $d_j^R \geq d_i^R$. Put $A = N_R(j)\setminus\{i\}$ and $B = N_R(i)\setminus\{j\}$. Removing a possible edge ij changes both sizes equally, so $|A| \geq |B|$. Since $v \in B \setminus A$, there is $k \in A \setminus B$. The four vertices are distinct, vi, jk are present, and vj, ik are absent. The switch $vi, jk \mapsto vj, ik$ adds a desired neighbor of v without losing an already correct one. Repeat and freeze v. The next residual degree vector depends only on the previous one and the

prescription, and remains graphical. With fewer than four remaining vertices a discrepancy would still force the four distinct vertices just exhibited, so none occurs. Every starting state reaches the same prescribed graph, proving connectivity.

Finally $\langle f, Hf\rangle = \sum_a \|h_a f\|^2$. Its vanishing forces f to be constant on each pair fiber. Every switch exchanges singleton assignments in a fiber indexed by either bipartition class of its toggled four-cycle. Connectivity therefore makes $\ker H$ exactly the constants. Positivity and (A.8) force every eigenvalue to be zero or at least one. On centered functions the quadratic form is at least the squared norm. Centering any f proves (2.1), including singleton state spaces.

References

1. Weibo Fu, Qian Qin, and Guanyang Wang, *Spectral gap for the binary fixed-margin swap chain*, arXiv:2606.22636v2, 2026, Version 2.
2. OpenAI, *Polynomial mixing of the switch chain for every graphical degree sequence*, OpenAI Math Release preprint, 2026, September 25, 2026; pinned commit 1581d4cbdd69a0faf34b2089614fe2a814316a6a.

Mathematical supplement

The proof supplements, exact certificate data and arithmetic programs referenced by filename are included in the attached mathematical supplement (ZIP). Open the attachment using a PDF reader with file-attachment support.